Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two ideal gases at absolute temperatures T_1 and T_2 are mixed. There is no loss of energy in this process. If n_1 and n_2 are the respective number of molecules of the gases, the temperature of the mixture will be

(A) $\frac{n_1T_1+n_2T_2}{n_1+n_2}$

(B) $\frac{n_2T_1+n_1T_2}{n_1+n_2}$

(C) $T_1+\frac{n_2}{n_1}T_2$

(D) $T_2+\frac{n_1}{n_2}T_1$

Q.2

A thermally insulated vessel contains an ideal gas of molecular mass M and ratio of specific heats γ. It is moving with speed v and it suddenly brought to rest. Assuming no heat is lost to the surroundings, its temperature increases by

(A) $\frac{(\gamma-1)}{2\gamma R}Mv^2K$

(B) $\frac{\gamma Mv^2}{2R}K$

(C) $\frac{(\gamma-1)}{2R}Mv^2K$

(D) $\frac{(\gamma-1)}{2(\gamma+1)R}Mv^2K$

Q.3

If 2 moles of an ideal monatomic gas at temperature T_0 is mixed with 4 moles of another ideal monatomic gas at temperature $2\ \mathrm{T}_0$, then the temperature of the mixture is

(A) $\frac{5}{3}\ \mathrm{T}_0$

(B) $\frac{3}{2}\ \mathrm{T}_0$

(C) $\frac{4}{3}\ \mathrm{T}_0$

(D) $\frac{5}{4}\ \mathrm{T}_0$

Q.4

The temperature of the mixture of one mole of helium and one mole of hydrogen is increased from $0^\circ\mathrm{C}$ to $100^\circ\mathrm{C}$ at constant pressure. The amount of heat delivered will be

(A) $600\mathrm{cal}$

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(B) 1200cal

(C) 1800cal

(D) 3600cal

Q.5

Three perfect gases at absolute temperatures T_1, T_2 and T_3 are mixed. The masses of molecules are m_1, m_2 and m_3 and the number of molecules are n_1, n_2 and n_3 respectively. Assuming no loss of energy, the final temperature of the mixture is:

(A) $\frac{n_1T_1+n_2T_2+n_3T_3}{n_1+n_2+n_3}$

(B) $\frac{n_1T_1^2+n_2T_2^2+n_3T_3^2}{n_1T_1+n_2T_2+n_3T_3}$

(C) $\frac{n_1^2T_1^2+n_2^2T_2^2+n_3^2T_3^2}{n_1T_1+n_2T_2+n_3T_3}$

(D) $\frac{(T_1+T_2+T_3)}{3}$

Q.6

The work of $146\ \text{kJ}$ is performed in order to compress one kilomole of gas adiabatically and in this process the temperature of the gas increases by 7°C. The gas is $(R = 8.3\ \text{Jmol}^{-1}\ \text{K}^{-1})$

(A) diatomic

(B) triatomic

(C) a mixture of monatomic and diatomic

(D) monatomic

Q.7

One litre of oxygen at a pressure of $1\ \text{atm}$, and 2 litres of nitrogen at a pressure of 0.5 atm are introduced in the vessel of 1 litre capacity, without any change in temperature. The
total pressure would be

(A) $1.5\ \text{atm}$

(B) $0.5\ \text{atm}$

(C) $2.0\ \text{atm}$

(D) $1.0\ \text{atm}$

Q.8

If the potential energy of a gas molecule is
$\text{U} = \text{M}/\text{r}^6 - \text{N}/\text{r}^{12}$, M and N being positive constants. Then
the potential energy at equilibrium must be

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(A) zero

(B) $\mathrm{M}^2/4\ \mathrm{N}$

(C) $\mathrm{N}^2/4\mathrm{M}$

(D) $\mathrm{MN}^2/4$

Q.9

A gas mixture consists of 2 moles of oxygen and 4 moles of Argon at temperature T. Neglecting all vibrational moles, the total internal energy of the system is $x\mathrm{RT}$. Then find the value of x

(A) 10

(B) 13

(C) 11

(D) 12

Q.10

Work done by a system under isothermal change from a volume V_1 to V_2 for a gas which obeys Vander Waal's equation $(V-\beta n)\left(P+\frac{\alpha n^2}{V}\right)=nRT$ is

(A) $nRT\log_e\left(\frac{V_2-n\beta}{V_1-n\beta}\right)+\alpha n^2\left(\frac{V_1-V_2}{V_1V_2}\right)$

(B) $nRT\log_{10}\left(\frac{V_2-n\beta}{V_1-n\beta}\right)+\alpha n^2\left(\frac{V_1-V_2}{V_1V_2}\right)$

(C) $nRT\log_e\left(\frac{V_2-n\beta}{V_1-n\beta}\right)+\beta n^2\left(\frac{V_1-V_2}{V_1V_2}\right)$

(D) $nRT\log_e\left(\frac{V_1-n\beta}{V_2-n\beta}\right)+\alpha n^2\left(\frac{V_1V_2}{V_1-V_2}\right)$

Q.11

Two moles of helium gas is mixed with three moles of hydrogen molecules (taken to be rigid). What is the molar specific heat of mixture at constant volume? $(R=8.3\ \mathrm{J/molK})$

(A) $21.6\ \mathrm{J/molK}$

(B) $19.7\ \mathrm{J/molK}$

(C) $15.7\ \mathrm{J/molK}$

(D) $17.4\ \mathrm{J/molK}$

Q.12

The volume of 2 moles of a diatomic ideal gas at $300\ \mathrm{K}$ is doubled keeping its pressure constant. Find the change in the internal energy of the gas. Given $R=8.3\mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$

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(A) 1.335 x 10^4 J

(B) 1.555 x 10^4 J

(C) 1.625 x 10^4 J

(D) 1.245 x 10^4 J

Q.13

One kg of a diatomic gas is at a pressure of $8 \times 10^4\ \mathrm{N/m^2}$. The density of the gas is $4\ \mathrm{kg/m^3}$. What is the energy of the gas due to its thermal motion?

(A) 5×10^4 J

(B) 6×10^4 J

(C) 7×10^4 J

(D) 3×10^4 J

Q.14

An enclosure of volume V contains a mixture of $8\ \mathrm{g}$ of oxygen, $14\ \mathrm{g}$ of nitrogen and $22\ \mathrm{g}$ of carbon dioxide at absolute temperature T. The pressure of the mixture of gases is (R is universal gas constant)

(A) $\frac{RT}{V}$

(B) $\frac{3RT}{2V}$

(C) $\frac{5RT}{4V}$

(D) $\frac{7RT}{5V}$

Q.15

H_2, O_2, N_2 and He are enclosed in identical containers under the similar conditions of pressure and temperature. The gases will have

(A) same R.M.S. speed

(B) same $\frac{K.E.}{kg}$

(C) different $\frac{K.E.}{\text{mole}}$

(D) same $\frac{K.E.}{\text{vol}}$

Q.16

Two thermally insulated vessels 1 and 2 are filled with air at temperature (T_1, T_2), volume (V_1, V_2) and pressure (P_1, P_2) respectively. If the valve joining the two vessels is opened, the temperature inside the vessel at equilibrium will be

(A) $T_1 + T_2$

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(B) $(T_1 + T_2)/2$

(C) $T_1T_2(P_1V_1 + P_2V_2)/(P_1V_1T_2 + P_2V_2T_1)$

(D) $T_1T_2(P_1V_1 + P_2V_2)/(P_1V_1T_1 + P_2V_2T_2)$

Q.17

Let γ denote the ratio of specific heat for an ideal gas. Then choose the correct expression for number of degrees of freedom of a molecule of the same gas.

(A) $\frac{25}{2}(\gamma - 1)$

(B) $\frac{3\gamma-1}{2\gamma-1}$

(C) $\frac{2}{\gamma-1}$

(D) $\frac{9}{2}(\gamma - 1)$

Q.18

Two identical calorimeters A and B contain an equal quantity of water at 20 °C. A 5 g piece of metal X of specific heat 0.2 cal g^{-1} $°C^{-1}$ is dropped into A and 5 g piece of metal Y is dropped into B. The equilibrium temperature in A is 22 °C and that in B is 23 °C. The initial temperature of both the metals was 40 °C. The specific heat of metal Y (in cal g^{-1} $°C^{-1}$) is

(A) $\frac{27}{85}$

(B) $\frac{54}{85}$

(C) $\frac{81}{85}$

(D) 0.4

Q.19

One mole of a diatomic gas undergoes a process $P = \frac{P_0}{\left[1 + (V/V_0)^3\right]}$, where P_0 and V_0 are constants. The translational kinetic energy of the gas when $V = V_0$ is given by

(A) $\frac{5P_0V_0}{4}$

(B) $\frac{3P_0V_0}{4}$

(C) $\frac{3P_0V_0}{2}$

(D) $\frac{5P_0V_0}{2}$

Q.20

Temperature at which the kinetic energy of gas molecule is half of the value of kinetic energy at 27°C is

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(A) $13.5^{\circ}\mathrm{C}$

(B) $150^{\circ}\mathrm{C}$

(C) $75\ \mathrm{K}$

(D) $-123^{\circ}\mathrm{C}$

Answers & Solutions

Q.1 Answer:

$\frac{n_1T_1+n_2T_2}{n_1+n_2}$

Solution:

Average kinetic energy per molecule of a perfect gas $= \frac{3}{2}kT$

$\therefore$ Average KE of molecules of the first gas

$= \frac{3}{2}n_1kT_1$

Average KE of molecules of the second gas

$= \frac{3}{2}n_2kT_2$

$\therefore$ Total KE of the molecules of the two gases before they are mixed is

$$K = \frac{3}{2}n_1kT_1 + \frac{3}{2}n_2kT_2$$

$$= \frac{3}{2}\left(n_1T_1 + n_2T_2\right)k \quad (1)$$

If T is the temperature of the mixture, the kinetic energy of the molecules $(n_1 + n_2)$ in the mixture is

$K' = \frac{3}{2}\left(n_1 + n_2\right)kT$

Since there is no loss of energy $K = K'$. Equating (1) and (2) we get

$n_1T_1 + n_2T_2 = \left(n_1 + n_2\right)T$ or $T = \frac{n_1T_1+n_2T_2}{n_1+n_2}$

Hence the correct choice is (a).

Q.2 Answer:

$\frac{(\gamma-1)}{2R}Mv^2K$

Solution:

As no heat is lost, Loss of kinetic energy = gain of internal energy of gas

$$\frac{1}{2}mv^2 = nC_V\Delta T \Rightarrow \frac{1}{2}mv^2 = \frac{m}{M}\cdot\frac{R}{\gamma-1}\Delta T$$

$$\Rightarrow \Delta T = \frac{Mv^2(\gamma-1)}{2R}K$$

Q.3 Answer:

$\frac{5}{3}\ \mathrm{T}_0$

Solution:

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Let T be the temperature of the mixture, then $\mathrm{U} = \mathrm{U}_1 + \mathrm{U}_2$

$\Rightarrow \quad \frac{\mathrm{f}}{2}\left(\mathrm{n}_1 + \mathrm{n}_2\right)\mathrm{RT}$

$= \frac{\mathrm{f}}{2}\left(\mathrm{n}_1\right)\left(\mathrm{R}\right)\left(\mathrm{T}_0\right) + \frac{\mathrm{f}}{2}\left(\mathrm{n}_2\right)\left(\mathrm{R}\right)\left(2\ \mathrm{T}_0\right)$

$\Rightarrow \quad (2+4)\mathrm{T} = 2\ \mathrm{T}_0 + 8\ \mathrm{T}_0\ (\because \mathrm{n}_1 = 2, \mathrm{n}_2 = 4)$

$\therefore\ \mathrm{T} = \frac{5}{3}\ \mathrm{T}_0$

Q.4 Answer:

$1200\mathrm{cal}$

Solution:

$(C_p)_{mix} = \frac{\mu_1 C_{p_1} + \mu_2 C_{p_2}}{\mu_1 + \mu_2}$

$\left(C_{p_1}(He) = \frac{5}{2}R \text{ and } C_{p_2}\left(H_2\right) = \frac{7}{2}R\right)$

$(C_p)_{\mathrm{mix}} == \frac{1\times\frac{5}{2}R + 1\times\frac{7}{2}R}{1+1} 3\mathrm{R} = 3 \times 2 = 6\mathrm{cal/mol.^\circ C}$

$\therefore$ **Amount of heat needed to raise the temperature from** $0^\circ\mathrm{C}$ **to** $100^\circ\mathrm{C}$

$(\Delta Q)_p = \mu C_p \Delta T = 2 \times 6 \times 100 = 1200\mathrm{cal}$

Q.5 Answer:

$\frac{n_1 T_1 + n_2 T_2 + n_3 T_3}{n_1 + n_2 + n_3}$

Solution:

Number of moles of first gas $= \frac{n_1}{N_A}$ **Number of moles of second gas** $= \frac{n_2}{N_A}$ **Number of moles of third gas** $= \frac{n_3}{N_A}$ **If there is no loss of energy then** $\mathrm{P}_1\ \mathrm{V}_1 + \mathrm{P}_2\ \mathrm{V}_2 + \mathrm{P}_3\ \mathrm{V}_3 = \mathrm{PV}$

$\frac{n_1}{N_A}RT_1 + \frac{n_2}{N_A}RT_2 + \frac{n_3}{N_A}RT_3 = \frac{n_1+n_2+n_3}{N_A}RT_{mix}$

$\Rightarrow T_{\mathrm{mix}} = \frac{n_1 T_1 + n_2 T_2 + n_3 T_3}{n_1 + n_2 + n_3}$

Q.6 Answer:

diatomic

Solution:

$W = \frac{nR\Delta T}{1-\gamma} \Rightarrow -146000 = \frac{1000\times 8.3\times 7}{1-\gamma}$

or $1 - \gamma = -\frac{58.1}{146} \Rightarrow \gamma = 1 + \frac{58.1}{146} = 1.4$

Hence the gas is diatomic.

Q.7 Answer:

$2.0\ \mathrm{atm}$

Solution:

According to Dalton's Law $\mathrm{P} = \mathrm{P}_1 + \mathrm{P}_2 \ldots\ldots\ldots\ldots\ldots\ldots\ldots$

here 1 litre of orygen at a pressure of $1\ \mathrm{atm}$, **&2 litre of nitrogen at a pressure of 5 atm are introduced in a vessel of 1 litre capacity.**

so $\mathrm{P} = \frac{\mathrm{V}_1\mathrm{P}_1 + \mathrm{P}_2\ \mathrm{V}_2}{\text{volume of vessel}} = \frac{1\times 1 + 2\times .5}{1} = 2\ \mathrm{atm}$

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Q.8 Answer:

$M^2/4\ N$

Solution:

$F = \frac{dU}{dr} = -\frac{d}{dr}\left[\frac{M}{r^3} - \frac{N}{R^{12}}\right] = -\left[\frac{-6M}{r^2} + \frac{12\ N}{r^{13}}\right]$

In equilibrium position, $F = 0$

$\therefore \frac{6M}{r^2} - \frac{12\ N}{r^{13}} = 0$ **or,** $r^6 = \frac{2\ N}{M}$

$\therefore$ **Potential energyat equilibrium position** $U = \frac{M}{(2\ N/M)} = \frac{N}{(2\ N/M)^2} = \frac{M^2}{2\ N} - \frac{M^2}{4\ N} = \frac{M^2}{4\ N}$

Q.9 Answer:

11

Solution:

Internal energy of 2 moles of oxygen $U_{O_2} = \mu\left(\frac{5}{2}RT\right) = 2 \cdot \frac{5}{2}RT = 5RT$

Internal energy of 4 moles of Argon. $U_{Ar} = \mu\left(\frac{3}{2}RT\right) = 4 \cdot \frac{3}{2}RT = 6RT$

$\therefore$ **Total internal energy** $U = U_{O_2} + U_{Ar} = 11RT$

Q.10 Answer:

$nRT\log_e\left(\frac{V_2 - n\beta}{V_1 - n\beta}\right) + \alpha n^2\left(\frac{V_1 - V_2}{V_1 V_2}\right)$

Solution:

According to given Vander Waal's equation $P = \frac{nRT}{V - n\beta} - \frac{\alpha n^2}{V^2}$

Work done,

$W = \int_{V_1}^{V_2} PdV = nRT\int_{V_1}^{V_2}\frac{dV}{V - n\beta} - \alpha n^2\int_{V_1}^{V_2}\frac{dV}{V^2}$

$= nRT\left[\log_e(V - n\beta)\right]_{V_1}^{V_2} + \alpha n^2\left[\frac{1}{V}\right]_{V_1}^{V_2}$

$= nRT\log_e\left(\frac{V_2 - n\beta}{V_1 - n\beta}\right) + \alpha n^2\left[\frac{V_1 - V_2}{V_1 V_2}\right]$

Q.11 Answer:

$17.4\ J/molK$

Solution:

C_{v1} **of helium** $= (3/2)R$

C_{v2} **of hydrogen** $= (5/2)R$

C_v **of mixture** $= \frac{2\times\frac{3}{2}R + 3\times\frac{5}{2}R}{(2+3)}$

C_v **of mixture** $= 17.4\ J/molK$

Q.12 Answer:

1.245 x 10^4 J

Solution:

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According to kinetic theory, there are no internal forces of interaction between the molecules of an ideal gas. This implies that the potential energy is zero. Hence, for an ideal gas, the internal energy is only due to kinetic energy of the molecules.

For n moles of an ideal gas at absolute temperature T, the internal energy is

$U = \frac{f}{2}nRT$

where f is the number of degrees of freedom. For a diatomic gas $f = 5$. Therefore

$U = \frac{5}{2}nRT$

Since pressure is kept constant, $V \propto T$ (Charles' law), i.e.

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$\Rightarrow \quad T_2 = \left(\frac{V_2}{V_1}\right) T_1 = 2T_1 = 600 \text{ K}$

$\therefore \quad (\because V_2 = 2V_1)$

$\Rightarrow \quad \Delta U = U_2 - U_1 = \frac{5}{2}nR\left(T_2 - T_1\right)$

$\therefore = \frac{5}{2} \times 2 \times 8.3 \times (600 - 300)$

$= 1.245 \times 10^4 \text{ J}$

Q.13 Answer:

$5 \times 10^4 \text{ J}$

Solution:

Volume $= \frac{\text{mass}}{\text{density}} = \frac{1}{4} \text{ m}^3$

$\text{K.E} = \frac{5}{2}\text{PV} = \frac{5}{2} \times 8 \times 10^4 \times \frac{1}{4} = 5 \times 10^4 \text{ J}$

Alternatively:

$\text{K.E} = \frac{5}{2}nRT = \frac{5}{2}\frac{m}{M}RT = \frac{5}{2}\frac{m}{M} \times \frac{PM}{d}$

$[\because PM = dRT]$

$= \frac{5}{2}\frac{mP}{d} = \frac{5}{2} \times \frac{1 \times 8 \times 10^4}{4} = 5 \times 10^4 \text{ J}$

Q.14 Answer:

$\frac{5RT}{4V}$

Solution:

The pressure exerted by a gas is given by

$$P = \frac{nRT}{V}$$

$$= \frac{\text{mass}}{\text{molecular weight}} \times \frac{RT}{V}$$

Pressure exerted by oxygen $P_1 = \frac{8}{32}\frac{RT}{V} = \frac{1}{4}\frac{RT}{V}$

Pressure exerted by oxygen $P_2 = \frac{14}{28}\frac{RT}{V} = \frac{1}{2}\frac{RT}{V}$

Pressure exerted by carbon dioxide

$P_3 = \frac{22}{44}\frac{RT}{V} = \frac{1}{2}\frac{RT}{V}$

From Dalton's law of partial pressures, the total pressure exerted by the mixture is given by

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$$P = P_1 + P_2 + P_3$$
$$= \frac{1}{4}\frac{RT}{V} + \frac{1}{2}\frac{RT}{V} + \frac{1}{2}\frac{RT}{V}$$
$= \frac{5RT}{4V}$, **which is choice (c).**

Q.15 Answer:

same $\frac{K.E.}{\text{vol}}$

Solution:

$\frac{K.E.}{\text{vol}} = \frac{3}{2}P$. **Here** P **is constant**

Q.16 Answer:

$T_1T_2\left(P_1V_1 + P_2V_2\right) / \left(P_1V_1T_2 + P_2V_2T_1\right)$

Solution:

According to the kinetic theory, the average kinetic energy (KE) **per molecule of a gas** $= \frac{3}{2}kT$. **Let** n_1 **and** n_2 **be the number of moles of air in vessels 1 and 2 respectively. Before mixing, the total** KE **of molecules in the two vessels is**

$$E_1 = \frac{3}{2}n_1kT_1 + \frac{3}{2}n_2kT_2$$
$$= \frac{3}{2}k\left(n_1T_1 + n_2T_2\right)$$

After mixing, the total KE **of molecules is**

$E_2 = \frac{3}{2}\left(n_1 + n_2\right)kT$

where T **is the temperature when equilibrium is established. Since there is no loss of energy (because the vessels are insulated),** $E_2 = E_1$ **or**

$\frac{3}{2}\left(n_1 + n_2\right)kT = \frac{3}{2}k\left(n_1T_1 + n_2T_2\right)$

or $T = \frac{n_1T_1 + n_2T_2}{(n_1 + n_2)}$

Now $P_1V_1 = n_1RT_1$ **and** $P_2V_2 = n_2RT_2$ **which give**

$n_1 = \frac{P_1V_1}{RT_1}$ and $n_2 = \frac{P_2V_2}{RT_2}$

Using these in Eq. (1) and simplifying, we get

$T = \frac{T_1T_2(P_1V_1 + P_2V_2)}{(P_1V_1T_2 + P_2V_2T_1)}$

Q.17 Answer:

$\frac{2}{\gamma - 1}$

Solution:

For, E **= Total Energy,** f **= Degree of freedom of molecule, We Know,**

C_v **= Molar heat capacity at constant volume** $= \frac{\partial U}{\partial T}_v = \frac{f}{2}R$

C_p**= Molar heat capacity at constant pressure** $= C_v + R = \frac{(f+2)}{2}R$

γ **= Ratio of specific heats** $= \frac{C_p}{C_v} = \frac{f+2}{f}$

$\Rightarrow f = \frac{2}{\gamma - 1}$

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Q.18 Answer:

$\frac{27}{85}$

Solution:

For X:

$5 \times 0.2 \times (40 - 22) = m \times C_{\text{water}} \times 22 - 20$

$\Rightarrow 9 = mC_{\text{water}}$

For Y:

Let C be the specific heat of the metal Y, then

$5 \times C \times (40 - 23) = m \times C_{\text{water}} \times (23 - 20)$

$\Rightarrow 85C = 27$

Therefore, $C = \frac{27}{85}$ cal g^{-1} $°C^{-1}$

Q.19 Answer:

$\frac{3P_0V_0}{4}$

Solution:

$P = \frac{P_0}{1 + V / {V_0}^3} = \frac{P_0}{2}$

$T = \frac{P_0V_0}{2R}$

Therefore, translational kinetic energy is equal to

$\frac{3}{2}RT = \frac{3R}{2}\frac{P_0V_0}{2R} = \frac{3P_0V_0}{4}$

Q.20 Answer:

$-123°C$

Solution:

Kinetic energy of a gas molecule

$E = \frac{3}{2}kT$

where k is Boltzmann's constant.

$\therefore \quad E \propto T$

or $\frac{E_1}{E_2} = \frac{T_1}{T_2}$ or $\frac{E}{(E/2)} = \frac{300}{T_2}$

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or $T_2 = 150$ K

$T_2 = 150 - 273 = -123$℃

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